Marmara University
Electrical and Electronics Engineering Department
Spring 2024
EE 4051 – Electromagnetic Waves
Simulation Project - Due March 22, 2024

Simulation of EM Wave Propagation

The main purpose of this simulation project is to numerically solve the time-domain electromagnetic wave equation. The EM wave propagation within a defined space of given properties is simulated. The finite-difference time-domain (FDTD) method is used to study EM wave propagation properties in two dimensions.

Introduction

The general form of the EM wave equation is given (for the electric field) by,

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \tag{1}$$

A similar equation can be written for the magnetic field. For a charge-free non-conducting space, the equation reduces to (in rectangular coordinates),

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \tag{2}$$

In general, the electric field is described by a vector of three-components as,

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z \tag{3}$$

Here, a simplified case is considered, where the electric field has only on component (E_z) varying only in x and y dimensions. Equation 2 becomes scalar, and is written as,

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)E = \mu\varepsilon\frac{\partial^2 E}{\partial t^2} \tag{4}$$

In this project, the Finite-Difference Time-Domain (FDTD) method will be used. The FDTD method is one of the most popular numerical methods in the field of computational electromagnetics [1, 2]. Its main advantages are (1) easy to formulate, (2) suitable for many structures (3) flexible for modifications and (4) robust.

The FDTD-based simulation is performed in the following steps.

1) Discretize the given structure (make a grid) using a suitable step size in both dimensions (Δx and Δy). The accuracy of the results improves with smaller step sizes.
2) Approximate the partial derivatives in equation 4 by the following relations.

$$\frac{\partial^2 E}{\partial x^2} = \frac{E(i+1,j) - 2E(i,j) + E(i-1,j)}{\Delta x^2} \tag{5}$$

$$\frac{\partial^2 E}{\partial y^2} = \frac{E(i,j+1) - 2E(i,j) + E(i,j-1)}{\Delta y^2} \tag{6}$$

$$\frac{\partial^2 E}{\partial t^2} = \frac{E^{n+1} - 2E^{n} + E^{n-1}}{\Delta t^2} \tag{7}$$

where i and j are the indices along the x and the y-axis, respectively. The index n is for time. Substituting equations 5, 6 and 7 in equation 4, and using $\Delta x^2 = \Delta y^2 = \Delta^2$, we get

$$E^{n+1}(i,j) = 2E^n(i,j) - E^{n-1}(i,j) + \frac{\Delta t^2}{\mu\varepsilon\,\Delta^2}[E^n(i+1,j) + E^n(i-1,j) + E^n(i,j+1) + E^n(i,j-1) - 4E^n(i,j)] \quad (8)$$

This is called the "update equation". It represents the time evolution of the wave at any given space point.

3) Implement the initial and boundary conditions and the material properties of the solution space.
4) Execute the time domain algorithm (equation 8) over all points in the specified space (excluding the boundaries) from $t = 0$ to any specified time duration (time marching).

Procedure

1) Consider the structure given in figure 1. The 2D space grid is generated covering the whole indicated space.

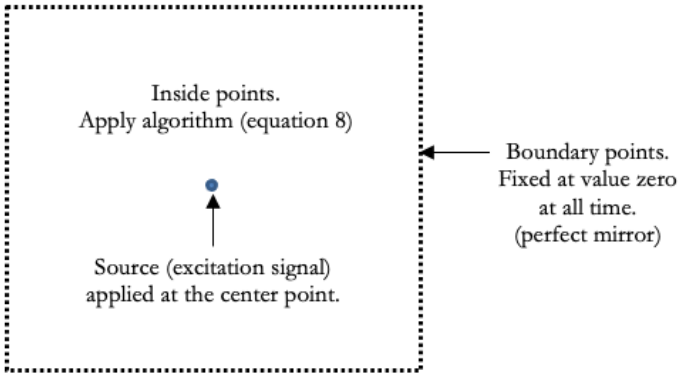


Fig. 1. Computational domain.

2) The boundary conditions are fixed at zero value throughout the simulation time. These conditions simulate a perfect mirror, where the EM wave is expected to reflect completely.
3) Implement the simulation algorithm for the inner point of the structure only. Apply a source (excitation signal) as an initial condition defined only at the center point. The applied signal is defined as a Gaussian pulse of the form,

$$E_s(t) = e^{-\left[\frac{t-t_o}{\delta}\right]^2} \quad (9)$$

where t_o is a time delay and δ is a parameter that controls the width of the pulse.
4) Find a suitable time step for the algorithm. A time step larger than a certain limit causes the algorithm to be unstable (diverges). As a rule, $\Delta t \leq 0.5\Delta/c$, for 2D problems.
5) Run the algorithm for the required time duration, observe the wave propagation and make studies and analyses.

Required Work

1) Consider the structure given in figure 1. The dimensions of the structure are 500 cm x 500 cm, and it is filled with free space. For a step size of 5 cm in both dimensions, generate the space grid.
2) The parameters for the excitation pulse are t_o=0.05 μs and δ=t_o/3.
3) Use MATLAB to implement the simulation algorithm. Make sure your program is concise, parametric, well-defined and well-documented.

4) For better visualization of the wave propagation, implement a "video" in MATLAB to inspect your results.
5) The expected velocity of the wave is c, the speed of light. So, let's use the velocity to test the accuracy of the results.
6) From the simulation, estimate the wave velocity and calculate the error.
7) Repeat for step sizes 2, 1 and 0.5 cm. Note that, in each case, the time step has to be adjusted. Perform error analysis (plot a curve).
8) Repeat the steps above for a simulation space that has ε_r=2.5. Calculate the wave speed in this case and compare to the expected speed. Give observations.
9) Study the wave propagation in the space given in figure 2. The boundary conditions and the excitation pulse are as before.

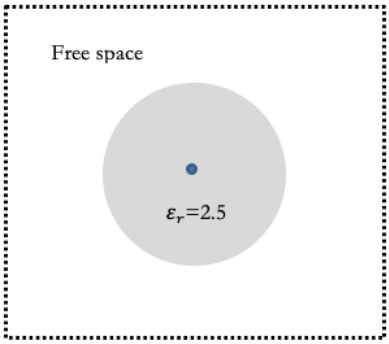


Fig. 2. Structure under study.

10) Study the propagation based on other interesting time-domain excitations (for example, sinusoidal, cosine-Gaussian pulse, etc.) Give observations and comments.
11) Study new interesting modifications to the given structure (unique to your team). Give all details with comments and observations.
12) (Bonus) Implement the simulation algorithm that includes medium loss (the term $\mu\sigma\frac{\partial \vec{E}}{\partial t}$ in equation 1). Show results and give comments.

Instructions

- Use MATLAB for solution and plotting.
- Up to 2 students can work together as a team.
- Submissions should be typed and in a formal report format. Include all analysis, codes, plots, calculations, observations, explanations and conclusions.
- Prepare your project report (one file) in pdf format; include everything within.
- In case of copied work (code, analysis, graphs, etc., even partially), all involved parties will get nulls.

[1] A. Taflove. *IEEE Trans. Electromagn. Compat.* 1980, **22**, pp 191–202.
[2] A. Taflove and S. C. Hagness. *Computational Electrodynamics: The Finite-Difference Time-Domain Method*, 3rd ed.; Artech House: Norwood, MA, 2005.

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